Matrix inequalities and norms

A universal block-norm characterization of essentially Hermitian matrices

Difficulty: challenging **Importance:** interesting to the community

Problem statement

Let $n \geq 1$ and $X \in \mathbb{C}^{n\times n}$. Suppose that for every pair of Hermitian $A, B \in \mathbb{C}^{n\times n}$ for which

$$H = \begin{bmatrix} A & X \\ X^* & B \end{bmatrix} \geq 0,$$

one has $\|H\|_2 \leq \|A + B\|_2$, where $\|\cdot\|_2$ is the operator norm. Must there exist a Hermitian K and scalars $\alpha, \beta \in \mathbb{C}$ with $X = \alpha K + \beta I_n$?

Such an X is called essentially Hermitian; equivalently its numerical range $\{v^* X v : \|v\|_2 = 1\}$ lies in an affine line.

Why it matters

The question characterizes exactly when off-diagonal coupling in a positive block matrix can always be ignored in a particular spectral-norm bound. It connects a numerical-range geometry condition to a universal norm estimate.

References

1. J.-C. Bourin and E.-Y. Lee, *Eigenvalue inequalities for positive block matrices with the inradius of the numerical range*, arXiv:2111.15180v1 (30 November 2021), Conjecture 3.3, Theorem 3.2, and Proposition 3.4. Primary text.
2. T. Hayashi, *On a norm inequality for a positive block-matrix*, Linear Algebra and its Applications 566 (2019), 86–97, Theorem 2.5. Preprint, DOI.

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The latest arXiv version of reference 1 remains v1. Hayashi proves normality under additional invertibility and distinct-singular-value assumptions, not the stated conclusion in full generality. A Frobenius-norm variant characterizes normality and is already proved. Searches included *Bourin essentially Hermitian conjecture*, *universal positive block matrix norm Hayashi conjecture*, and *essentially Hermitian conjecture proof 2025 2026*. No later resolution of Conjecture 3.3 was located. A 2025 result on decomposable numerical ranges settles a different conjecture.